

Optimized Sensor Position Detection: Improving Visual Sensor Setups for Hand Tracking in VR

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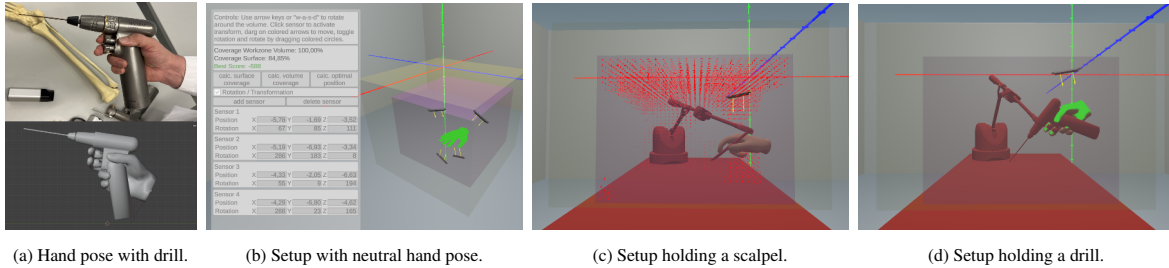
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(a) Hand pose with drill.

(b) Setup with neutral hand pose.

(c) Setup holding a scalpel.

(d) Setup holding a drill.

Figure 1: Hand poses were modeled after reference images from surgeons (a). Sensors visualized as black rectangles with arrows showing their orientation. Light purple volume: Hand movement area. Light yellow volume: Limit for sensor positions. Objects that could potentially block the view of sensors are in red. Voxels that are not covered by the sensor view are visualized as red crosses (c), hand surface that is covered is visualized in green (b, d).

ABSTRACT

Hand tracking plays an important role in many Virtual Reality (VR) applications, enabling natural user interactions. Achieving precise tracking is often challenged by occlusion and suboptimal sensor placement. To address these challenges, we developed the *Sensor Positioning Simulator*, a versatile tool designed to optimize sensor placement. To demonstrate its utility, we simulated scenes from the VIRTOSHA project, a VR-based surgical training platform. Evaluations show that the tool effectively positioned sensors to achieve maximum hand surface visibility and full hand movement area coverage, even in occlusion-heavy environments. Future developments include support for animated simulations and validation through real-world experiments.

Index Terms: Hand Tracking, Visual Sensors, Sensor Positioning, Simulation

1 INTRODUCTION

Hand tracking has become a fundamental technology in VR applications, allowing users to interact naturally and intuitively within virtual environments without the need for physical controllers [1]. Among the various types of sensors available for hand tracking, optical sensors are particularly prevalent in VR applications due to their accessibility and ease of integration. To achieve high precision, sensors must be strategically placed to maximize coverage of the hands, while minimizing blind spots caused by occlusion or interference from other objects [2].

The diversity of VR applications, ranging from casual gaming to professional training, means that no single sensor configuration can meet the needs of every setup. VR developers are thus faced with the task of customizing sensor arrangements for their specific use cases, often through trial and error [3].

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This work introduces the *Sensor Positioning Simulator*, a tool designed to assist VR developers in optimizing their sensor setup. The simulator enables users to experiment with different sensor placements and evaluate their effectiveness through visualizations of hand-surface (Figure 1d) and areal coverage (Figure 1c). The tool features a *Sensor Position Finder* to automate the process of finding good sensor configurations. To demonstrate its practicability, we integrate the simulator into the VIRTOSHA project, a VR-based surgical training platform, showcasing its ability to address the complex tracking requirements of a highly specialized use case.

2 DESIGN

The overarching goal is to create a modular system, thus, several aspects of the simulator are made customizable. Each algorithm was implemented to work independently of specific meshes or environmental configurations. Users can adjust parameters, including the number of sensors and modify the range and field of view (FOV) to represent different sensor technologies. Further customization is available through defining the objects that are used in the setup, including obstacles that might occlude the sensors' views.

The *Sensor Positioning Simulator* supports two usages: sandbox mode and the *Sensor Position Finder*. In the sandbox mode, users can experiment with different sensor configurations. The effectiveness of a configuration can be assessed through integrated calculations of the hand's surface coverage and the volume in which the hand can move. The *Sensor Position Finder* provides an automated method for identifying optimized sensor configurations. The algorithm computes a configuration that maximizes surface and volume coverage. The results can be used by VR developers in their real-world setups, ensuring optimal hand tracking performance.

3 IMPLEMENTATION

The system was implemented in Unity 2022.3.41f1.

Volume coverage is calculated by subdividing the target volume into discrete units (voxels). To determine coverage, raycasting is used. Rays are cast from the sensors toward the volume, while the range and FOV of the sensor is considered. Voxels intersected by these rays are marked as "covered." The proportion of covered voxels v_{covered} relative to the total number of voxels v_{total} provides the volume coverage metric $V = v_{\text{covered}}/v_{\text{total}}$.

Table 1: Left columns: Results from the cross-validation of the *Sensor Position Finder*. Right columns: Parameters used in cross-validation.

Cross-Validation Results			Parameter					
Scenario	VolumeCoverage	SurfaceCoverage	Sensor		Environment	Sensor Position Finder		
Neutral	84% $\pm$ 3%	99% $\pm$ 2%	FOV	170x170°	Resolution Raycast	2000	α	0.5
Drill	77% $\pm$ 5%	98% $\pm$ 2%	Range	10-75cm	Resolution Voxels	1000	Initial Temperature	1000
Scalpel	74% $\pm$ 4%	99% $\pm$ 1%	Number of sensors	4	Resolution Mesh	818	Cooling Rate	0.99

Surface coverage involves calculating the extent to which a sensor can "see" the surface of the target object (hand). This is done by using raycasting, too: If a ray intersects with a face of the target mesh, this face is evaluated as "seen". The surface coverage $S = f_{\text{covered}}/f_{\text{total}}$ is calculated as the ratio of the mesh's hit faces f_{covered} to its total face count f_{total} .

The **Sensor Position Finder** uses a Simulated Annealing algorithm. In an iterative optimization process, each sensor position and orientation configuration (state vector $\vec{x}$) is gradually refined. First, a random configuration is picked for each active sensor. Neighboring solutions are created by changing the configuration of one sensor at a time. Each solution is evaluated based on an objective function that balances the surface and volume coverage metrics, ensuring neither is excessively prioritized at the expense of the other. In addition to the surface and volume coverage terms, we introduce a balancing term that applies a penalty when there is a large difference between the two values. This penalty is weighted by the balancing factor α .

$$\vec{x}_{\text{best}} = \underset{\vec{x}}{\operatorname{argmax}} - (S(\vec{x}) - 100)^2 - (V(\vec{x}) - 100)^2 - \alpha(S(\vec{x}) - V(\vec{x}))^2$$

4 USE CASE

The *Sensor Positioning Simulator* was developed as a solution to address the specific requirements of the VIRTOSHA project¹ that aims to create a VR-based surgical training environment that incorporates haptic feedback. The system enables trainees to practice fine motor skills and surgical techniques in a safe, controlled setting. Using a haptic robot arm, the resistance encountered when performing tasks like drilling into bone or cutting soft tissue are replicated. For the training to be effective, the system demands extremely precise hand and object tracking.

To address this challenge, two representative surgical scenes were recreated in the *Sensor Positioning Simulator*, one involving the use of a scalpel (see Figure 1c) and the other involving a drill (see Figure 1d). These scenes include the environment modeled after the workspace, complete with a desk, robotic arm, and surgical instruments. Using reference images provided by surgeons, we recreated the required hand poses. The models of the hand, tools, and environment are loaded into the simulator scene to optimize sensor positioning for this specific use case. The project uses four ultraleap Stereo IR 170 sensors (specifications see Table 1).

To evaluate the effectiveness of the simulator and the position-finding algorithm, we cross-validated the computed outcome. In three conditions (neutral hand position, drill use, scalpel use, see Figure 1) sensor optimization was performed 10 times. The performance of the algorithm was assessed based on its ability to position sensors effectively and its overall coverage metrics. The parameters used in the setup are stated in Table 1.

4.1 Results

The *Sensor Position Finder* consistently positioned the four sensors to minimize the impact of occlusion. For areas prone to self-occlusion or occlusion by blocking objects, the algorithm ensured that these regions were visible from multiple angles. The results of

the cross-validation show that the algorithm found configurations with very high volume coverage and high surface coverage in all three conditions (see Table 1).

5 LIMITATIONS AND FUTURE WORK

One of the primary challenges in using the simulator is balancing the voxel and mesh resolution with the number of rays cast by the sensors. If the resolutions are high, computational demands increase, and the simulation may become inefficient. Moreover, insufficient ray density can lead to inaccurate results, as not all voxels or mesh faces may be hit. For instance, if mesh resolution and rays are imbalanced with too few rays relative to the mesh detail, many faces will remain undetected, leading to an underestimation of coverage. Conversely, if the voxel or mesh resolution is too low, the system may overestimate coverage.

While the current version of the *Sensor Positioning Simulator* provides the essential functionality, upcoming developments will focus on further enhancing adaptability and real-world relevance. Planned features include animated objects to simulate hand movement and dynamic obstacles, adapting the *Sensor Position Finder* to work with moving objects. We will adapt this new feature by recording hand poses and movements during surgery and create an animated simulation of our use case.

To further evaluate the *Sensor Position Finder*, we plan to experimentally validate the simulation outcomes and their real-world performance. This involves replicating sensor configurations identified by the simulator in a real-world setup and assessing their effectiveness in achieving optimal hand tracking in VR.

6 CONCLUSION

The *Sensor Positioning Simulator* offers an approach to solve the challenge of sensor placement for hand tracking in VR applications. By providing a flexible, customizable simulation environment, the tool allows developers to fine-tune sensor configurations and maximize tracking coverage. As VR continues to evolve and hand tracking becomes even more integral, tools like this simulator will play an important role in improving the design and functionality of VR systems. Our future work will focus on further developing the tool and validating its real-world performance.

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